

ActInf Textbook Group ~ Cohort 4 ~ Meeting 13 (Chapter 6 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 13 (Chapter 6 part 1) | 52:13

all right welcome back everyone we're in
cohort 4 of active inference textbook
group and picking up on the second round
of
activity over the last few months in
cohort 4 some of you were there there
were some people who were not there who
are here now and vice
versa we went through the first five
chapters and now in this last three
months of of
2023 we're going to head through the
second half of the
book so before we like go into anything
even
remotely active conceptual does anyone
just want to share like where did the
first half of the book leave
them Andor what are they excited about
in the second half of the book for this
interval
or any
questions that are kind of guiding
people's curiosity and learning
Journeys I thought that this was the
first meeting for cohort 5 but
I'm it works for me yeah just just
briefly to that note
[Music]
um it's every Tuesday there's a cohort
five and a cohort four and they're like
an alternating times so sometimes people
just join whichever time is more
convenient for them but either way it's
all recorded it's all here it's all the

same textbook and it's many Cates of
paint so there's like literally no
reason to even read one chapter before
another in fact chapter six is a very
great entry point in certain
ways excellent I'll probably come to
both anyway cool Smail and then anyone
else who wants to raise their
hand my my question is about our Discord
group uh long time I'm looking for uh
some mass group for learning but I don't
know um I can't find uh there is some
group about mathematic in the Discord or
no now is stopped
um so you're referencing the math
learning group there's no special place
in Discord this this math learning group
it just kind of like it's like a team
name that if there's people who are
motivated to like pick up that banner
and commit to doing some organizing like
say I will be here weekly to do this and
that or here's where we're going to take
notes people are welcome to do that so
it's just kind of like there for people
who would love to see it be there
but we're not doing anything further and
it's not like there's like another Forum
or anything like that everybody who
registered is added to all the calendar
events and this is the shared document
for
everything yeah
thanks but yeah where does the first
half of the book leave us and then how
do we set an intention or what is
exciting about the second half of the
book yeah mail or anyone

else uh I thought the first half was pretty comprehensive um and I do I like your description of many coats of paint because it is like like each chapter you are going over a lot of things that you covered in a different way in the previous chapter but in a sort of new light

um and I do Wonder like you know how that will come back up like a lot of the concepts will come back up when we're sort of Designing the models which I think like the next half is more focused on like actual

applications

um

and I'd say you know there are certain things I want to like go back over on myself just later in the previous chapters especially like message passing and more of the neurobiology stuff which I uh am very interested in but um I'm not as familiar as like maybe just some of the regular like beian statistics aspects of

it cool thank

you yeah again anyone can raise their hand or write anything

just to recap though the first five chapters the author say is the epistemic or the learning part and that's where the kind of what is active inference is covered first there's a context provided then the low and the high road are gone through low road is how that's basian statistics the high road is why that's that persistence

imperative chapter four goes into the

generative models of active inference
which exactly what the second half of
the book is about applying however
chapter 4 gives a pretty long and
comprehensive treatment of what a
generative model is and then chapter
five studies the setting where the most
generative models have been constructed
and deployed and associated with
empirical data which is mammalian
neuroscience so it's a very tight first
half of the book and then we get into
the second half of the book and in the
beginning of chapter
six they call their shot they said this
chapter provides a four-step recipe to
construct an active inference model and
it discusses a bunch of other design
choices the quote here give me six hours
to chop down a tree and I will spend the
first four sharpening the axe pretty
interesting it's like okay well how much
time do you have to make this active
inference generative model do you have
10 minutes and you're scribbling it on a
napkin or do you have four years or 15
years to develop this and then here we
are in this
chapter where it's like yeah what is
sharpening the axe what is the tree that
we're going to cut
down kind of starts it on an
interesting note
um is there anything anyone wants to
like ask or say about just active
inference generative
modeling
we can dip into chapter six now and look

at the recipe

itself but is there anything people are thinking or curious about in terms of like active inference generative modeling it's an exciting moment and this in certain ways is the topic itself in

practice all of the discussion around an active inference model of this and of the icade and of depression and of organizations and of Gaia like those are active inference generative models of insert system of Interest here so that's what the earlier part of the textbook approaches in

theory and chapter six is where the recipe is laid out and there's a lot of dots to connect and some of it we can add detail and connective

tissue but this is kind of like the real moment where it active inference goes from being like something that's being discussed on stage or conceptualized by others in their mind into being something that we enact through

constructing a generative model like reading the first five chapters to continue with this culinary analogy is like reading recipe books or reading a Molecular Gastronomy textbook or reading about micro and macro nutrients or the myard reaction or like all these like sort of like ideas about cooking and here's the recipe and then there's still more guide books and playby plays and workplace training and all these kinds of things would be associated with the practice of cooking but this is where

that culinary textbook does a crossover moment into talking about the recipe itself thank you Andrea yes we can definitely overview six also on the top of each chapter page there there is a short textual o a video with a short overview that olly and I have prepared for for the chapters um okay so let's look at the the recipe itself

so section one they're just talking section one of chapter six is an introduction and they do make some very relevant points and then in section 6.2 we see the recipe itself so we've pulled the recipe out just so we can kind of expand and

discuss the recipe asks four questions which system are we are we modeling what is the most most appropriate form for the generative model how to set up the generative model and how to set up the generative process and then they write these four steps in most cases suffice to design an active inference model once completed the behavior of the system is determined by the standard schemes of active inference specifically or technically

The Descent of the active in internal States the autonomous states on the free energy functional associated with the model

so the recipe is asking us questions that only we can address situationally that help us set up the landscape could be a very Dynamic landscape it might be a very trivial one

that just kind of rolls to a single Valley but these questions are going to help us construct the landscape that then the ball rolls downhill on so it's kind of like making a pinball machine or like making some other kind of um rub Goldberg machine we're at the second level conceptualizing of a system and then asking that that's like what what system are we modeling what do we want to model and it uh isn't necessarily always uh easy which system we're modeling someone is joining we'll see um but but this question about system of interest is in common to active inference generative modeling as well as to just broader systems engineering and we get into the active inference specific questions here about the appropriate form of the generative model so this is the first this is the first of the three practical challenges and this is brought up in chapter 4 earlier when discussing gener models in discrete or continuous time and then it's revisited in chapters seven and 8 which are going to go into more detail on time but it's one of the big upfront questions that come into play after the system of interest has been determined is where and how time is going to be traded maybe if anyone wants to like what's a system of interest that we

could kind of carry through for the rest of this

discussion as an example or just to kind of ground some of these more General topics like an example of something like um example of what we would want to model with this recipe yeah like what would you answer to the first question um

could be like you know a robot that we're trying to just model using U active

inference sounds good let's continue with the robot

setting what is the most appropriate form for the generative model so that brings up this time question

and you could imagine different time conceptions maybe even different sensors and actuators on the robot have a different time

conception from that early

question steps three and four are really where the work

happens the second question with time is an early modeling

decision that sets up

broadly which branch chapter seven or chapter eight the model is going to be built in line with these are kind of

like two architectural styles with their Associated technical differences

um as they write if you have a process that's more in terms of categorical

inferences that leads you to the discret

case so like chess the moves in the

board are discret in time there's like

a Click Of Time anywhere that there's a

click of
time discrete time is usually
appropriate where there's a flow or an
unfolding of time then the continuous
time models are specialized on this and
then as they point out you can make
Hybrid models so a really common example
of a hybrid model has at the lower or
the sensory motor level continuous
models and then at the higher cognitive
level something more
discrete but that question about
time sets up all of the work
itself and this is
where really the iterated modeling comes
out so we're we're dealing with this
robot and um we're going to make a
discrete Time model because it's a
digital computer and so this discrete
formulation represents
well how this robot works we could zoom
out and make a continuous time model
with a with a person or we could zoom in
and um make a continuous model for some
sensor or actuator but just at the robot
level think about discrete time so
then that just sets
up really the canvas for all of these
questions
what are the generative models most
appropriate variables and
priors which parts are fixed and must
must be
learned um if anyone wants to like add
anything please like raise your hand or
give a thought but
but there's there's little else we can
say like this is the work itself of the

modeling after you've kind of unrolled the canvas and decided um that we're going to be talking about a discrete time model these are all of the real modeling

questions that basically take you from I'm interested in a robot and I'm going to do a discrete Time model all the way on through a description of a system that's necessary and sufficient for understanding its

Behavior does anyone have like thoughts or ideas on this

sorry the last thing you said you uh were muted there for a second just does anyone have a thought or question on

this well I think it's it's pretty interesting uh one thing that stuck out to me not from the textbook but from listening to the last panel in the posium was what Bert de re was talking about you know needing to be able to scale models according like say for an autonomous let's say the robot is an autonomous uh self-driving car and it needs to be able to to calculate with a different degree of precision whether it's driving you know straight down a freeway or going around a roundabout and so I guess would that be an example of you know trying to build a a hybrid you know a hybrid model that based off that circumstances is you know following a different route in that pathway in the tree it's a great connection so let's

we're we're in the room talking about
the generative model of the self-driving
car yeah and now we're thinking about
two possible ways we could
we could have like a fixed Precision
model or we could have a variable
Precision
model even if somebody doesn't have a
lot of background in modeling it's
pretty clear like what the strengths and
weaknesses would be well in the fixed
Precision model it's simpler and if you
fix it at an adequate frequency or
Precision it might work totally fine
okay and what do you get with a variable
Precision model well you have this
hyperparameter which is like how
variable or how should I change how I
change the Precision so it's not like
you've answered the last mile by saying
we'll make it variable Precision you
could imagine a variable Precision model
that just adds hyperparameters that
don't make it a more effective robot or
take up a huge amount more computational
resources so that's exactly the
conversation about which features of the
model do we want to
combine yes please Jesse uh could you
give an example of like in that case
what what would be an example of a hyper
parameter that you
might yeah programing yeah in that
specific case the
hyperparameter would so the the the the
parameter itself would be how
precise the fixed parameter model would
just say Precision level 10 you could

think of that as the function y equals
10 yeah it's just a static line and then
the
hyperparameter would be now say well
let's do inference on what that
Precision parameter should
be and you might say I'm going to draw
precisions from a distribution that has
this and that shape you want to draw
precisions so let's just say the fixed
example is 10 do we want to draw our
precisions from 10 plus or minus two or
10 plus or minus 8 or do we want to
sweep between 10 and 100 whatever
strategy you take in a basian setting is
like the hyper prior that's you kind of
surveying out or mapping
out another space that you explore the
parameters of the Distribution on and
the double-edged sword is almost always
quite literally hyperparameters are more
expressive and they can fit situations
better
however they introduce
compounding exponential search
spaces so if you just have to do
something you could get into a lot of
recursive especially in cognitive
modeling a lot of
recursive
possibilities that are like way past the
point of diminishing return of being
relevant basian model selection
is a principled way to understand how
many parameters to include in a model
and and so on so there are methods to
address this question of like how many
parameters should we have how meta do we

have to go and those are exactly the kinds of evaluative criteria that would come into play again we're in the room talking about the two different generative models fixed Precision variable Precision or time Horizon of three and time Horizon of 12 are we getting a better model period with a longer time Horizon is it worth it those are exactly the kinds of questions that come into like model evaluation so that would be like the cooking evaluation stage here's the recipe and the questions that you ask when you're the chef preparing this recipe and then there could be another chapter on how do you evaluate cuisine in general or for some specific genre yeah Jesse um with the limiting variable on that generally be computational power or just just uninterpretable results both great great suggestions yeah in principle you you can always add more hyperparameters you might be making a bigger search space and just sifting through a progressively larger space to find little needles in the Haystack and then exactly the second piece which is like the interpretability and AI interpretability and all these things are very important and sometimes a parameter can be constructed to have a certain semantics or

interpretability other times there are parameters that you could kind of describe but parameters don't need to necessarily be interpretable however one of the cool things about active inference especially because we're taking such a Hands-On approach to building these generative models is we can Design interpretable Systems and that's something that Ines Hippolito and colleagues have recently explored in a little bit more detail as well but it's exactly that interpretability that we're like in the Farmers Market Boutique generative model construction Workshop here talking about what's the game plan for how we're going to make this Artisan dinner that's kind of where chapter six sits

okay and maybe if at some point you could put in the chat maybe a link to like a paper in respect to that the interpreted what you're just saying via interpret

interpretability or maybe in the textbook in theoda y this page I see you right here but yeah I'll I'll add it to the page totally good call all right thank you um yeah from June of this Sor for that aside oh no it's it's it's good I mean this is like we want to be Hands-On with the models so that we can understand them and um this is how the the textbook shows some simple settings so from earlier in the book we had the frog that did or

didn't jump out of the person's hand or the object that was in their hands that did or didn't jump and that was a kind of case study in passive inference the person's just holding the object it jumps or it doesn't and then they're going to update their beliefs about whether it's a frog or an Apple so that brings up a lot of important topics like bayesian belief updating and learning surprise bayesian surprise censor Fe multimodal inference but it's passive in the second half of the book a kind of canonical active inference setting is going to be the rat in the te-as that's a setting where the world doesn't just happen to the animal it's like its actions matter for what sensory stimuli it gets and so this example of like a teamas where there's something that's rewarding or preferred at the ends or either or both end and then there's something informational at the bottom of the te-as that's kind of like a minimal model system for active inference because it captures different things that you might want to emphasize like the value of information or habit all these different features so these are kind of like pedagogical or toy examples that are very very graspable and very small scale models and then in the literature you see more like medium to large scale

models or really more in the small to medium space and then in applications you might see Large Scale Models but from the research and development side oftentimes people are just like I introduced a new Motif and made a small scale model that demonstrates The Motif and then they move on on developing motifs because if you can develop the motif and make it Rock Solid in a small case that's actually clearer and more preferable and then people can introduce that Motif into medium and large scale models a lot more to say about modeling but but yeah anyone can like give a thought or a question about modeling yeah I guess in that respect I'm just curious about you know model switching and if there I guess are if there's Master models that can determine when to switch between models in a given you know based off updating you know updating either a belief system or or essentially learning your Environ learning an environmental circumstance yeah lot to say on that um let's look at figure 4.3 this is our Rosetta Stone figure on the top is the discrete time generative model on the bottom is the continuous time generative model they're being positioned provocatively

to highlight their structural similarities there's the downstairs part that looks like an e and that's like the passive part going through time and then on the top part in the upstairs of each there's the policy selection action policy selections being treated a little bit differently between these two settings and what the world is is treated differently between the two settings in the discrete time setting we have a past past present and a future time point that emit observations and then you have policies that intervene into how the world changes through time and thus which observations it emits but in the discrete time setting you have a past a present and a future you could have it look 100 time steps in the past and one in the future or none in the past and 100 in the future those are the kinds of parameters that the modeler plays with whereas in the continuous time model you have only the present and then you have derivatives of the present like the first second third derivative that reflect how the present is changing through time so we have like the car's location is at location 100 and it's going 10 miles per hour so that means at the next time point we expect it to be at 110 now if we said well it's at 100 and its speed is 10 and its acceleration is one so then it's going to get to a little bit more than 110 by the next time point in continuous time but you're

all at every moment you can like recheck
the flow okay so you asked about model
switch so let's look to ahead a bit to
figure um

8.6 so here the upstairs part is a
discrete model we know we're in the
discrete setting because we have like
the past present and future so that's
the discrete time partially observable
Markov decision
process and then
coming out of each
observation we have like another little
figure

4.3 and it's the bottom part of figure

4.3 with the continuous time

model so you could compose discrete on
discrete continuous on continuous and so
on but in this case let's think about
model switching there's a world where
there's two possibilities either the
weight is 5 lb or the weight is 50 lb
and so you have some belief before
anything else happens about like How
likely is it to be five versus 50 and
then there's a descending
expectation like I'm in
five that then is going to set a
policy on the sensory motor
model and so you could imagine that like
if you expected
five and then it was five or if you
expected 50 and it was 50 it would be
not surprising that's like when the step
is at the distance that you expect the
step to be those are like literally the
things that we don't attend to when our
descending predictions are

accurate however if there was a mismatch
there was a prediction
error that some that it was heavier than
you
thought that would update your belief at
the higher level in the way you'd expect
and vice versa
so descending
predictions can one example or one like
way we could think about those is like
those are the models that you can switch
over and then that sets the top- down
expectation which includes action
policies because we're not just
expecting sensory data we're expecting
ourselves to do things that's why we select
the most likely thing because the top
down expectations are over the sense and
the
action and
then when what we get back from the
ascending levels is what we expect it's
the least surprising which means that we
had the most model
evidence the prediction ER was the
lowest the surprise was the lowest the
free energy was the lowest things are
good the ball is on the bottom of the
hill and then as sensory feedback
diverges you have more
surprise that means the ball is not on
the bottom of the bowl there's a higher
free energy and so
on sorry oh yeah uh in that respect
could you consider prediction error
almost as like a thermostat for model
switching
where

it could be used as such if you made the model that had that kind of a feature or or attribute than it literally would be if you didn't it wouldn't be and this is what and that's what Carl Friston and others sometimes call a deflationary account which is like deflating all the hot air out of the account and just ending up with the lean distilled you know if and only if necessary and sufficient whole truth nothing but the truth status that's the deflationary account he like well I made a cognitive model where ascending prediction are used as a cue for model switching then the deed is done but until the generative models on the table the discussion is very speculative it's like until you have on the table the linear aggression connecting smoke and cancer you might have a super strong anecdotal belief you might have a strong prior but if you're just speculating especially if you think well depending on how you include covariates it's not so simple before you make the model even for very large effect phenomena in the world which is why statistics is a process and why it matters so chapter six shows us the recipe and is like hey there is a bright line between like the recipe talking about the food and having the food on the plate in front of you the recipe is how you get from being like I would like to have this dish to a

dish your version of the dish isn't
necessarily going to look like Food
Channel

status but it will be what you did
do and so that is really the clear line
between Ultra speculative
hypotheticals like what does active
inference say about attention there's
not even a system of interest in that
sentence realistically that's not even a
complete

question because you could depending on
the system of interest do any number of
things just too many to even say in one
sentence but we have a recipe and a
process that actually works towards
giving a deflationary
account

so here is one case where uh Michael
Carl I believe we looked a little bit
about like translation and ey gaze so
humans Translating that could even be
understood as a a translation or
switching
behavior

[Music]

um
great point that several have raised
like the steps in the recipe don't need
to be sequential or separate so in a way
these are kind of like these are
everything you need to think about when
um you want to serve dinner but
depending on different people's kitchen
and setup like they might do one thing
before another now there's some things
that you might suffer if you do one
before the

other but there's other things things
where the order doesn't really matter
but just in a book or in a verbal speech
you have to present one in front of
another but just because things
are presented in a order doesn't mean
that that's like from most to least
important or least to most important any
of that so these are like just
questions that we ask oursel as part of
the iterated modeling
process and they're big questions that
AR aren't going
away and none of them are too simple to
speed over what system are we modeling
well it's like is it going to be the
robot and the niche but then where do
you go with the niche does that include
the power plant I mean where do you
really develop these reasonable
bounds and that's again to give a
deflationary account it's like what is
this model
for so you can never even from the very
outset disentangle the generative model
from it being specifically constructed
by a team in a
place and that's like a kind of
radical scientific inactiv ISM perhaps
or scientific
realism and again it's really
fascinating how there is a clear line
between generative model on the table
maybe limping along
maybe simple but generative model on the
table exhibiting simulated Behavior
being used to process
data and then we have like the space of

the cognitive it's so
productive but that's the conversation
around the dinner
table and if all the chefs walk away
from the table
and there's no food on the table maybe
it was just a conversation group group
but if the business plan was to produce
a certain kind of food with
reliability than just speculating about
variance to the recipe wouldn't have
gotten you
there then there's a lot of just big
topics that come up repeatedly like
having multiple time scales of
inference and like real systems are
happening all scales all times at once
and this is a fact that's often pointed
to for example with biological systems
but for a given system of interest or
phenomena of
interest there may be some spatial
temporal scales that are more relevant
to model like if we're simulating I
movement
Behavior if we make something that is is
iterating every Pico second we're just
way over
sampling and so it end there'd be a lot
of boring time steps and then there'd be
some shocking time steps and we would
have like a really inadequate model
conversely if we were to grain over
hours then we would think there was
nothing interesting about Gay's Behavior
because it ended up being just kind of
sampled looking no different than a
inanimate gas particle but then there's

certain observations certain scales of interest that have these rich cognitive Dynamics and the meaning isn't just simply like out there ready for us to pick it up it's actually in the co-construction of the model and in this analysis and the unfolding of a system of Interest one other useful figure from chapter nine they describe this as the metabasian perspective so here we have figure 4.3 in the center generative model discrete time generative model past present future so this is going to be the subjective model which is to say that of the subject of the experiment and then a little bit more philosophically or metaphysically the subject model could be that which is subjectively experienced but putting that aside Anil not sure if you're actually here or not but it'd be very cool if you were um this model is of the experimental subject and then here we're out in the room now so inside was the rat in the Maze that's our experimental subject but then from the perspective of the researcher the subject is the object and the behavior that it emits is objective Behavior so just like if we were doing an animal Behavior Experiment you'd have like well what does the animal know what is its input stimuli and what actions can it do what are its affordances so

that's like the system of interest and
then now what else is in the
room and then if you can develop that
thin
wrapper around your system of interest
you have the
minimal
setup to have something on the table and
maybe it's good enough with just that
first framing or maybe you want to go
way way way deeper in a certain way or
you want to explore some other cognitive
phenomena but broadly just connecting it
to behavioral sciences and just like
looking at like what is the animal under
observation and it's a cognitive model
and then what's the thin wrapper around
that that helps bring a closure to this
encounter so that we can get a grasp on
it

a uh I didn't have a question okay
okay so I I I think it's super cool
because a lot of these conversations
aren't necessarily active inference
specific spatial and temporal scale
system of Interest
identification um positioning of your
model multimotor Sensor
Fusion um
like in certain ways the active
inference
part in chapter six it's
like that's the work again all the part
that gets us to the kitchen are General
modeling
commentaries and then we have the
specific work itself which is setting up
the generative

model and then they just have a very
very brief section
6.7 which gets Revisited in um chapter
nine about interfacing data to these
generative
models so there's a lot of ways to go in
this chap chap is
like the recipe for
it what what what are anyone's thoughts
or or questions on
this yeah so I was wondering like I know
like in the active inference Institute
they have a lot of projects and sort of
uh G repositories where they show
examples of using these to model like
cognitive agents and things like that
but I was wondering like does this
textbook itself have sort of maybe like
a GitHub repo with examples of sort of
them using these recipes or things like
that um great great question very
important question in appendix C they do
provide some mat lab code we have our
own repo for the textbook so like we
might just want to bring that code
in and also develop other examples like
I would love to see
the chef and the Sue chef and and the
client everyone's notes and just
everyone's thoughts about like what
could be developed a lot of times people
pick really ambitious systems of
interest to
model and that can be like daunting
because these are you can go arbitrarily
sophisticated cognitive modeling and
three
humbling and often seen as like like not

the real work to build these simpler pedagogical examples to get a feel for these models by hand and like kind of start with lighter weights basically so that people can develop an intuition for these different tools and skills and practices

um seeing examples is helpful especially really simple ones seeing complex examples may actually not give the kind of information that people are always looking for because for really well-built examples the kernel is kind of lost in the specifics so if you have something that's taking in these five different kinds of sensors and doing this kind of a planning on this kind it's like the real essence of active inference is spread across thousands of lines of code and then you have to abstract away from those five different sensors and specific action setting to kind of reconstruct what your own carrying on board sense of active inference so the small and the medium models are really important and then that helps you Scope whether a large scale model is even possible so don't be discouraged yeah that that that the that the large scale model is not in hand that um you know that you're not a Factory Kitchen

today because you got curious about the
cuisine of
cooking maybe we could have a whole
whole whole restaurant
uh analogy
thread well I got an analogy for you on
that so to go off topic a little bit but
um out here there's like tons of prickly
peirs they probably have them up in
Davis too right they right now so I just
went and collected like a whole bucket
full and I've been making like U juice
from it but my hands are just so full of
thorns so I think I feel the same way
with active inference I'm trying to just
make a cup of juice and I've like got
Thorns all over my body
now that's funny yeah
just just like right at the very
beginning we're discussing like chapter
six I feel like chapter 1 and chapter 10
are a great entry points into the um
theory in the context of active
inference chapter six is like the sort
of doing out because chapter six grounds
the fulcrum of this
book in a almost
boringly inactive
framework
that's the deflationary
angle and uh it's the real work of
building the generative model not
everyone's going to build every
generative
model but it's super exciting that the
bright line exists that the culinary
sector
exists and that actually dancing amongst

different metaphors and systems of
interests we're able to have the
conversations we want to have so like
for systems designed
these questions from chapter 6 these are
things maybe you don't call it a
generative model or generative
process but like what are you interested
in what matters to you about it what
phenomena do you think are relevant like
those are questions that everyone can be
addressing and then to bring those
questions and awarenesses
legibly into a modeling process that can
have the Integrity of all other
engineering it's super exciting and it's
what among other things differentiates
acadm from being just like a hot take or
like a kind of mic
drop it's like yeah chapters 1 two 3
four five the mic is dropped
repeatedly and then what do we
do
cool
well we're in the game now cohort 4
we're
back we out there we back um we'll
continue in chapter six earlier time uh
slot next week and we'll be in the game
and and let's think about how we can
kind of Cate and improve our Niche for
future cohorts also maybe develop along
the weeks if someone has like a system
of interest that they want to kind of
develop out and play with different
features of like if you had a system of
Interest robot and you kind of answered
the four questions to the recipe and

then in week uh three and four in
chapter seven you explored the discrete
time model and then in the next weeks we
explored a continuous time feature of
the model and then in chapter nine you
thought about how data comes into play
and in chapter 10
recontextualized you know you'd have a
transformative understanding and trail
of work for a very limited amount of
time regarding your system of Interest
so if you choose something that cares uh
that you care about then that kind of
work sequence should be as meaningful as
anything could
be what time is the earlier time
um uh it will be uh it's at 13 UTC all
right 13 and I feel like there's going
to be some time zone changes or
something like that but uh like or
daylight savings or something
but um yeah it's all on the calendar
it's all also written here
so thank you all I hope that you
have a great rest of your day thank you
likewise thank you so much Daniel yeah
peace bye peace